

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VI

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Aim : Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

Code:

Manifest.json

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

Service-worker.js

```
self.addEventListener("install", e => {
  console.log("Service Worker Installed");
});

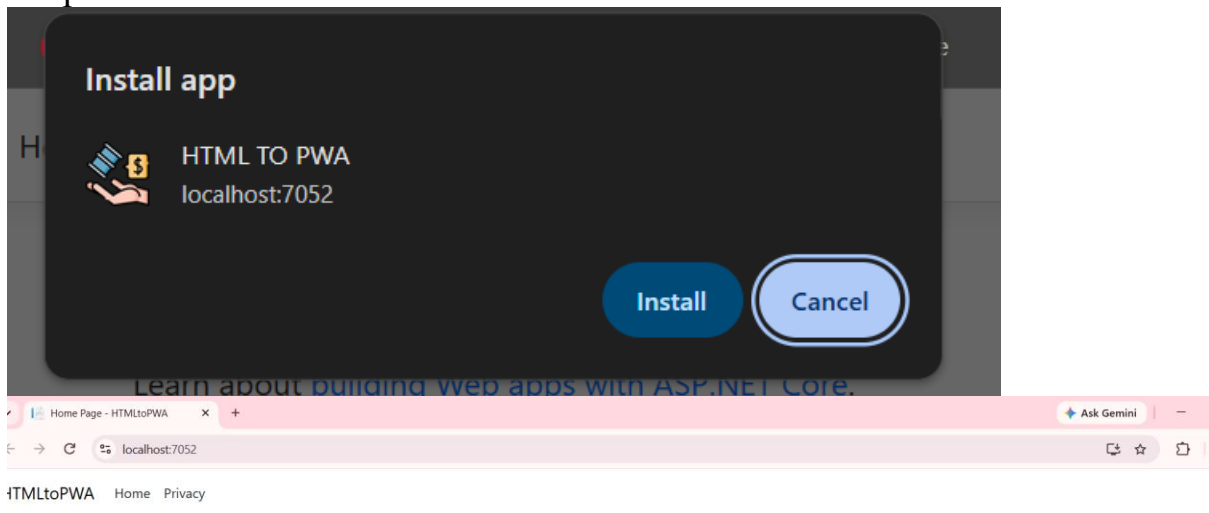
self.addEventListener("fetch", e => {
  e.respondWith(fetch(e.request));
});
```

```

<link rel="stylesheet" href="~/HTMLtoPWA.styles.css" asp-append-version="true">
<link rel="manifest" href="~/manifest.json" />
</head>
<script>
    if ('serviceWorker' in navigator) {
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>

```

Output:



College information portal

Welcome

Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:

Service-worker.js

```

const CACHE = "v1";

self.addEventListener("install", e => {
    e.waitUntil(caches.open(CACHE).then(cache =>
        return cache.addAll([
            "/",
            "/offline.html"
        ]));
    console.log("Service Worker Installed");
});

```

Offline.html

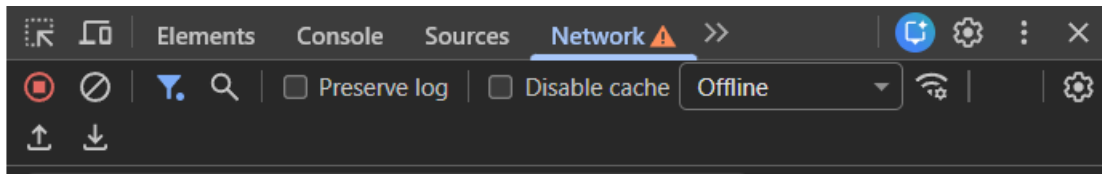
```

<meta charset="utf-8" />
<title>offline </title>
</head>
<body>
    <h2>please check your internet connection</h2>
</body>
</html>

```

Output:

please check your internet connection



Part C – Curiosity Question

Your college has a website that students use every day to check notices, timetables, and study materials.

Many students say:

"It would be easier if this website worked like a mobile app."

The college does not want to spend money on developing a separate Android or iPhone app.

Task

How can you convert the college website into an app-like experience that students can install on their phones?

Hint: Use a **Progressive Web Application (PWA)**.